



Roll No.	Name of the Candidate		Signature of the Invigilator	
22Z208	AKSHARAA P.		[Signature]	
Branch & Semester	BE CSE	8	Test No. 1	Date 27/1/26
Course Code	19Z011	Title MULTI TIER COMPUTING		

Faculty Use	Marks Scored					Grand Total (Out of.....)	Faculty Signature
	Qn.	a	b	c	Total		
	1	3	5	4	9	21	[Signature]
	2	4	2	2	8	18	

INSTRUCTIONS TO CANDIDATES

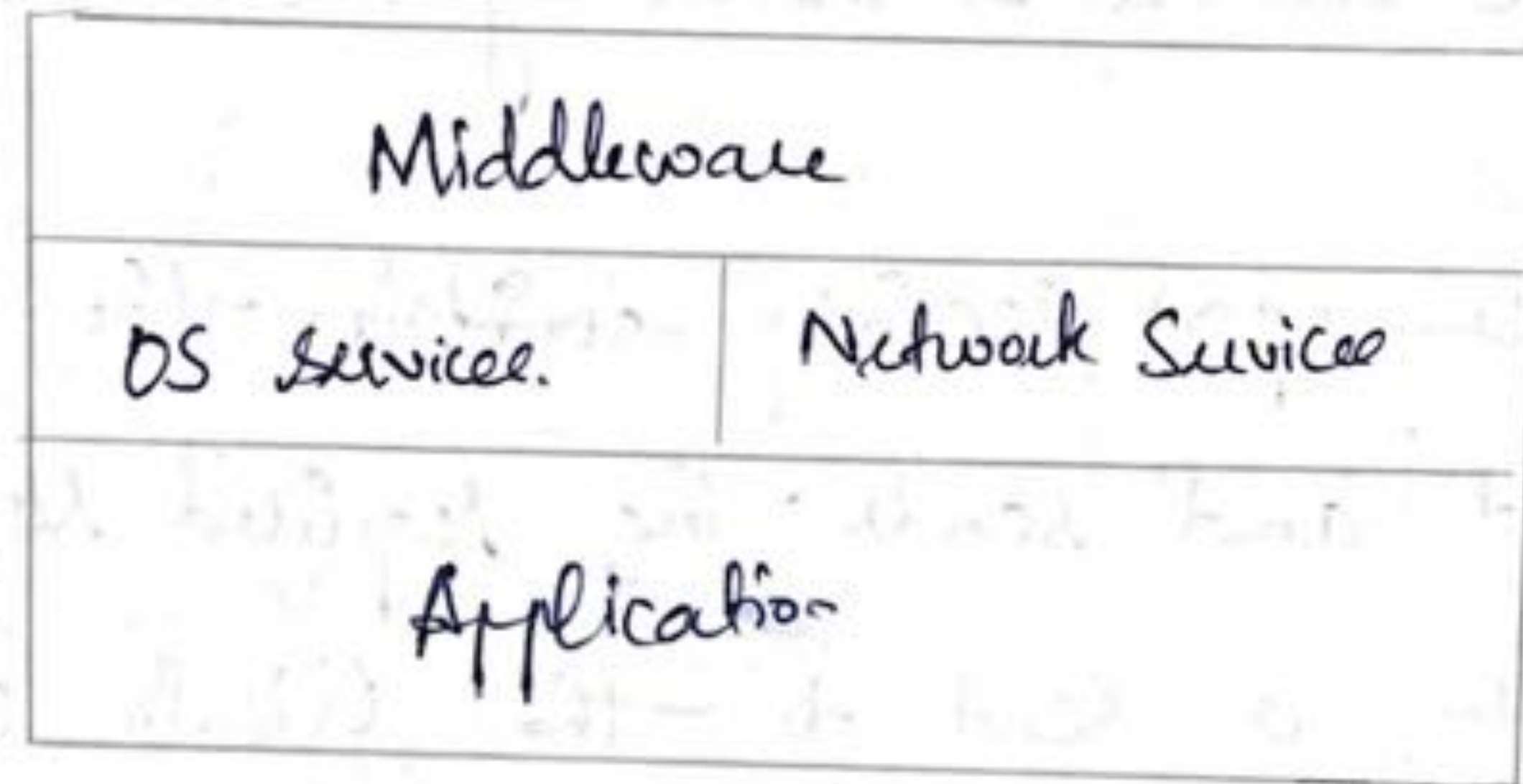
- The candidates shall not possess any **handwritten / printed materials / any incriminating materials / Cell Phones / Programmable Calculators / Smart Watches / Calculator Covers** inside the test hall. Possession of any materials / gadgets mentioned above inside the test hall will be treated as malpractice.
- Malpractice will be viewed very seriously and the **punishment** may be **invalidation of test in one or more courses**.
- No additional sheets** will be issued. After completing the test, the candidates shall personally handover the answer booklet to the Hall Superintendent.

I have read all the instructions above

- I understand and abide by the norms prescribed by the college during assessments.
- I have neither offered nor received any aid in answering the questions of this test.

[Signature]
Signature of the Candidate

- ii) * Middleware is a layer that lies between the Client and Server. It acts as an intermediate communication between Client and Server. The Middleware layer contains components like TP Monitor, ORB etc.



- * Middlewares only exist in a 3-tier Architecture
- * In a client-server interaction, the Client makes a request that is processed by the Middleware and then sends the request to the Server.
- * Middleware acts as the Business Logic where it tells the Client what the Server is responding and the Server about the Client's Request.



* The Client sends a request, which the Middleware processes and sends it to the Respective Server.

* Middleware monitors every transaction that is sent and received and the status of the transaction. The Transaction is written a destination server by the Middleware.

* The Server upon receiving ~~monitors~~ the transaction, processes the request and sends the required response.

✓ The Response is sent to the Client's output display and is viewed from there.

(c)

Key Components:

i) Client:

* Client involves the user and the UI.

* Clients make a request through the GUI.

* Client side is also where the final output is displayed (ie) Response to the Request.

* The Client gets a failure message if the transaction failed or got lost, the Client can re-request the transaction.

ii) Server:

i) Data Server:

* In Data Server, the data is stored in structured, unstructured and semi-structured in the Database Management System.

* Data Server is responsible for the CRUD operations

- Create,
- Read
- Update
- Delete.

* Data Retrieval, Data Updation, Data Deletion is done by the Data Server.

* SQL databases: MySQL, PostgreSQL

* NoSQL databases: MongoDB, Cassandra.

ii) Network Server:

* Network Server oversees the Transaction processing.

* It uses Techniques such as - TCP/IP

- Web Sockets for the transaction process

It ensures that the transaction is successful if not it takes countermeasures to inform the Client about the failure of the transaction.

(ii) Application Server:

In Application Server, it is connected to multiple data server and is responsible for the flow of the transaction, it monitors every step of the transaction.

(iii) Security:

Security protocols are maintained to ensure that no third party breaches the system and manipulates the data on the transaction, Some of the Network Security Risk include Interruption, Interception, Eavesdropping & Fabrication.

(iv) Load Balancing and Load Clustering:

The Workload of the data system is equally distributed to the server. This ensures that each task is prioritised and separately distributed among the servers.

v) Directory Services:

* DNS (Directory Name Services) and LDAP are used in the system.

vi) Resource Allocation:

Under utilization and Overutilization of resources to servers is avoided and prevented using Resource Allocation Methods.

2B) i)

The Anatomy of a Server,

- i) Initialization
- ii) Receive Request
- iii) Processing Request
- iv) ~~Business logic~~
- ~~iv) Request loading~~
- iv) Business logic
Request loading
- vi) Sending Response.

2bii) GUI based and OOU.

* In Object Oriented User Interface, everything is stored in a class or in an object.

* When a request needs to be sent, the class must be called or the object itself can be called so that it is sent through the Middleware layer.

* The GUI-based system is simpler in comparison to Object Oriented UI.

* However using OOU has shown lesser no of failures compared to GUI.

* This is due to the structural components of each.

* OOU supports features such as Inheritance of Classes, Encapsulation of Class and Data, It also supports abstraction.

* Whereas in GUI, there is no abstraction.

* When a transaction is being processed all data related is seen which also increases the Risk of Third party use attack.

2c)

Extended Operating Services:

i) Fault Tolerance:

- * Incorporating ^{Error} correction and Error detection techniques can increase the success rate and reduce the loss rate.
- * This way the Client doesn't encounter any failure message or has to go through the process of sending the request again.
- * Improves Customer Satisfaction.

ii) Transaction Management:

- * Multiple transactions are put across the server, this can overload the system and disrupt the overall working performance.
- * Managing Transaction based on priorities will also improve the client's satisfaction as it ensures top priority based tasks are done first.

iii) Prioritization:

- * There are many transactions being loaded, the server is so overloaded with tasks, at times immediately required transaction processing ends up at last.

- * To sort - this certain transaction are marked with priority, so that highly important transaction are processed first followed by the others.

iv) Load balancing and Load Clustering.

- * Combining activities of (ie transactions) together ensures a smooth and efficient overall quality.

v) Resource Allocation

- * Allocating equal resources to every server is not efficient as certain servers require less or more.
- * To prevent overuse and underuse of resources, Resource allocation methods are used so that required amount is allocated to each server's workload.

vi) Application Distribution and Directory Services

- * Distributing transactions across servers using DNS or LDAP.